

Asymmetric Capital Control: Executive Summary

Codes in parentheses refer to results in the proofs document, which carries an index of them. Claims not so marked are either framing or are attributed to the literature.

Two banks facing identical regulation can run visibly different capital policies, and the difference matters: one that treats a shortfall against its own target as far worse than an equivalent **surplus** is good will hold more capital, pay out later, and recapitalise sooner than the rules alone require. Whether that conservatism comes from the regulation or from the bank's own asymmetric attitude to shortfalls determines how it responds when the rules change—and the two are not distinguishable by inspection, because both produce a bank holding more capital than the minimum. Recovering the preference separately from the regulation is therefore the prior question, and this paper's central finding is that the asymmetric aversion is invisible in the natural **volatility-based measure** of it, and visible instead in the thresholds the bank commits to. In a solved model of a bank managing its **capital buffer** under a **regulatory cap**, the **definitional evaluation** of the **deficit-to-surplus volatility ratio**, computed directly from the model's structure rather than estimated, depends only on **cap geometry**—how the **regulatory cap** varies with the **buffer**—and carries no information about the preference at all (SCG). The preference does show up in where the bank commits to act, and both responses are obtained in closed form: a more shortfall-averse bank retains a larger **buffer** before paying out at the **dividend barrier**, and recapitalises earlier, from a higher level of capital, at the **recapitalisation trigger** (TCS). The **fixed issuance cost** moves that trigger the opposite way (KCS), and the opposition of the two is decisive: it renders the Jacobian of the two thresholds in the two parameters non-singular, so preference and cost are jointly identified from the boundaries alone (IDN). Much of empirical finance reads a firm's risk attitude off its volatility; in this model that **identification** fails outright, and the decision boundaries carry the information instead.

Two structural facts underpin that finding. First, setting the asymmetry parameter to its neutral value removes the **shortfall penalty** entirely, giving an **exact nesting** of the **risk-neutral benchmark** rather than a limiting approximation of it (NST), so the parameter measures asymmetry in excess of what regulation and costs alone would produce. Second, because the **shortfall penalty** is kinked-linear rather than S-shaped, the running payoff stays concave for every degree of asymmetry (RPC). This is the paper's departure from **reference-dependent** models built on the S-shaped **value function** of prospect theory (Kahneman and Tversky, 1979), which are convex below the **reference** and therefore require a concavification argument before any threshold structure can be recovered at all. The claim is stated at exactly that strength and no further: it concerns the objective's flow term, and does not deliver a concave **value function**. The **value function** is in fact locally convex above the **recapitalisation trigger** (LNC), but that non-concavity is caused by the **fixed issuance cost** and is present even at the neutral parameter value, so it is localised and cost-driven rather than global and preference-driven—precisely the difference from the S-shaped case.

A **discrete-time model** is worked out first and closed completely, with the **capital buffer** written as a ratio of equity to **required capital**. The **continuous-time formulation** restates

the same bank’s problem with three controls of distinct type: the **classical control** of **risky exposure**, constrained by the **regulatory cap**; the **singular control** of **dividends**, acting at the **dividend barrier**; and the **impulse control** of **recapitalisation**, carrying both a fixed and a **proportional issuance cost**.

The **regulatory cap** is not a single constraint but two regimes, a solvency branch and a liquidity branch, with the binding one switching at the level where they cross. The bank’s **capped exposure** is the smaller of the two at each point, which gives the cap a kink of its own, distinct from the kink in the preference. A consistency condition on the parameters governs behaviour at the **distress boundary**: the drift there is positive only if the return on assets exceeds the return on liabilities, which the parameters used throughout satisfy (CCP). This is what pushes the **buffer** away from distress rather than into it, and it is the reason the distress boundary is never reached under the solved policy.

An **accounting identity** ties the bank’s **return on equity** to its growth and retention (ACI). **Deficits** are taken to decay more slowly than **surpluses**, a **persistence asymmetry** (PSA), and from that the central discrete-time result follows: the **volatility ratio** is a fixed power of the asymmetry parameter (FPV), depending only on the **saturation limits** of the transition between the two states and not on how the transition is shaped in between (ROB). The same recursion is observationally equivalent to a standard **asymmetric volatility specification** of the EGARCH class (EGE), so the ratio could in principle be recovered from returns without observing **required capital** directly, and a first-passage argument gives the sojourn time—the expected length of time the **buffer** spends below the **reference**—in closed form (SJT).

In continuous time, a **liquidation payoff** bounds the **value function** from below (LQB), and a change of variable shows that the discrete-time state and the continuous-time **capital buffer** are one state measured from one origin (TSO). The apparent breakdown of the model at the distress boundary, the point at which the **buffer** exactly matches **required capital**, turns out to be a **coordinate artefact** rather than a feature of the problem: in the reconciled coordinate the model is well-behaved everywhere, and the boundary is never actually reached (CDA). That reconciliation is what makes it meaningful to compare the same preference’s legibility in the thresholds and in realised volatility on equal footing.

A separate finding concerns where the **regulatory cap** actually binds. Under the solved policy the cap is active across the whole **inaction region**: the solver searches the exposure over a grid spanning the admissible range and retains the maximiser, so a slack optimum would have been found had one existed, and at almost every solved case none is (CBI). What closes the problem is therefore the **impulse control** rather than any interior relaxation of the cap. The paper states this at the strength it has—a finding at these parameters rather than a theorem, with the formulation accommodating either outcome without modification—and the solver producing it is separately documented and validated (SLV).

The thresholds’ **comparative statics** give the fuller picture, and they are theorems rather than observations. Because the asymmetry parameter enters the objective linearly and never restricts the admissible set, the sensitivity of value to it is the expected discounted shortfall incurred under the optimal policy, and every boundary response reduces to a property of that single object (ENV). Both smooth-fit conditions then supply its boundary data, and the derivatives follow in closed form: $dy^*/d\lambda_S$ from the second-order fit at the barrier — the first-order condition being degenerate there — and $dx_L/d\lambda_S$ from a first-order expansion at the hitting time (TCS). The **fixed issuance cost** admits the identical treatment with the discounted count of injections in place of the shortfall, and the signs oppose on the trigger: a steeper penalty makes the bank intervene earlier, a larger fixed cost makes it wait longer, while both raise the barrier (KCS). That opposition is what makes the two parameters separately recoverable, and it does so unconditionally rather than by comparing

magnitudes (IDN). Empirically the two write themselves into different observables — preference moves the levels, the **fixed issuance cost** moves the **trigger-to-target gap**, which vanishes exactly at zero fixed cost (KSW). That threshold geometry should carry preference information is not without precedent at the firm level—Angelis (2025) derives an inaction band with two direction-dependent targets from loss aversion in a price-setting problem—but that model is a discrete behavioural one rather than a continuous-time control problem, and does not run the inverse mapping from band to preference.

What the results rest on is stated rather than assumed. Differentiability of the value in each parameter needs no hypothesis at all: the value is a supremum of functions affine in that parameter, hence convex in it, hence differentiable off a countable set (REG). Differentiability of the boundaries reduces to the implicit function theorem, whose non-degeneracy conditions turn out to be the same ones that sign the derivatives, so each does double duty (BDR). The two second-order conditions the identification needs are likewise theorems: because the impulse operator is affine, the **injection target** is independent of where the injection is made from, and the required curvature at each boundary follows from a sign-change argument and from the variational inequality on the intervention region (SOC). Verification itself is now unconditional: it rests on four hypotheses, all of which hold, and concavity is not among them — its absence is therefore no obstruction (VER). The one point that had remained — whether the **value function** is continuously differentiable at the **recapitalisation trigger** — is settled by the same variational inequality: a corner there would have to be convex, and a convex corner is incompatible with the value function being a viscosity supersolution, so smooth fit holds and the trigger derivative is continuous (SMF). What the numerics could not resolve, because the discrete scheme carries a corner at the boundary by construction, the argument settles by showing a corner cannot exist. Each of the document’s thirty-five numbered results carries one canonical verifier, recorded in its verification appendix: five are proved in Lean with no unproved gaps, four more have their algebraic cores machine-checked in Lean with the analytic input stated as an explicit hypothesis, twelve are exact symbolic checks, and the remainder are proofs in the text or labelled numerical findings. Separately, whether the **induced variance asymmetry**, as opposed to its **definitional evaluation**, moves with the **asymmetry parameter** is weakly identified rather than absent, and the paper reports the magnitude rather than claiming invariance.

Beyond those, the direction the finding opens is one of scope. The mechanism rests on a specific choice: the **kinked-linear penalty**, which keeps the **running payoff** concave (RPC) and, by the same feature, keeps the thresholds responsive to the preference. Whether committed thresholds remain informative under other classes of **shortfall penalty**—and which classes preserve the contrast between uninformative variances and informative boundaries—would establish how far the result extends past the case solved here. Armstrong and Brigo (2019) suggests the question has a sharp answer: their irrelevance results for constraints require a disutility that is sublinear in the left tail, a hypothesis a **kinked-linear penalty** does not satisfy, which is consistent with limits binding, and therefore revealing, in exactly this case. Establishing that connection is future work and not a result of this paper. To our knowledge no existing firm-level model combines singular payout control with impulse **recapitalisation** under an asymmetric preference rather than a constraint; the recent literature generalising preferences in dividend control remains singular-only (Chen et al., 2026; Cao et al., 2026). What this paper settles is which quantities in a solved bank’s policy carry the preference and which cannot, and the answer holds whether or not any bank is ever measured.

References

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